

Robotics 2

Adaptive Trajectory Control

Prof. Alessandro De Luca

DIPARTIMENTO DI INGEGNERIA INFORMATICA
AUTOMATICA E GESTIONALE ANTONIO RUBERTI



SAPIENZA
UNIVERSITÀ DI ROMA

Motivation and approach



- need of adaptation in robot motion control laws
  - large uncertainty on the robot dynamic parameters
  - poor knowledge of the inertial payload
- main tool and methodology
  - linear parametrization of robot dynamics
  - nonlinear control law of the dynamic type (the controller has its own 'state')
  - online update of estimates of dynamic coefficients in the model
- characteristics of direct adaptive control
  - direct aim is to bring to zero the trajectory error, i.e., position and velocity errors (state error)
  - estimates need not to converge to the true values of the dynamic coefficients of the robot (as opposed to indirect adaptive control)

Summary of robot parameters



- parameters assumed to be known
  - kinematic description based, e.g., on Denavit–Hartenberg parameters ($\{\alpha_i, d_i, a_i, i = 1, \dots, n\}$ in case of all revolute joints), thus including link lengths (kinematic calibration), as well as gravity acceleration g_0
- uncertain parameters that can be identified offline
  - masses m_i, positions r_{ci} of CoMs, and inertia matrices I_i of each link, appearing in combinations (dynamic coefficients) $\Rightarrow r \ll 10 \times n$
- parameters that are (slowly) varying during operation
  - viscous B_{vi}, Coulomb B_{ci}, static B_{si} friction at joints $\Rightarrow (1 \div 3) \times n$
- unknown (and abruptly) changing parameters
  - mass, CoM, inertia matrix of the payload (w.r.t. the tool center point)



when a payload is firmly attached to the robot E-E, only the 10 parameters of the last link are modified, influencing however most part of the robot dynamics

Goal of adaptive control



- given a twice-differentiable desired joint trajectory $q_d(t)$
  - with known desired velocity $\dot{q}_d(t)$ and acceleration $\ddot{q}_d(t)$
  - possibly obtained by kinematic inversion + joint interpolation
- execute this trajectory under large dynamic uncertainties
  - with a trajectory tracking error vanishing asymptotically

$$e = q_d - q \longrightarrow 0 \qquad \dot{e} = \dot{q}_d - \dot{q} \longrightarrow 0$$

  - guaranteeing global stability, no matter how far are the initial estimates of the unknown/uncertain parameters from their true values and how large is the initial trajectory error
- identification is not of particular concern: in general, the estimates of dynamic coefficients will not converge to the true ones!
- if this convergence is a specific extra requirement, then one should use (more complex) indirect adaptive schemes

Linear parametrization

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) + B_v\dot{q} = \tau$$

- there exists always a vector $\rho \in \mathbb{R}^r$ of dynamic coefficients, so that the robot model takes the linear parametrized form

$$Y(q,\dot{q},\ddot{q})\,\rho = \tau$$

- vector ρ contains only unknown or uncertain coefficients
- each component of ρ is in general a combination of the robot physical parameters (not necessarily all of them)
- model regressor matrix Y depends linearly on $\ddot{q}$, quadratically on $\dot{q}$ (only for the terms related to kinetic energy, not for friction), and nonlinearly (trigonometrically) on q
- same approach can be pursued, though much less efficiently, by using the linear parametrization in terms of (standard) dynamic parameters

$$Y_\pi(q,\dot{q},\ddot{q})\,\pi = \tau \qquad \pi \in \mathbb{R}^p \text{ (or } \in \mathbb{R}^{p_r}) \qquad \boxed{p \gg p_r > r}$$

Trajectory controllers
based on model estimates



- inverse dynamics feedforward (FFW) + PD (linear) control

$$\tau = \underbrace{\hat{M}(q_d)\ddot{q}_d + \hat{C}(q_d,\dot{q}_d)\dot{q}_d + \hat{g}(q_d) + \hat{B}_v\dot{q}_d}_{\hat{\tau}_d} + K_P e + K_D \dot{e}$$

$$\hat{M}, \hat{C}, \hat{g}, \hat{B}_v \quad \Longleftrightarrow \quad \text{estimates } \hat{\rho}$$

- (nonlinear) control based on feedback linearization (FBL)

$$\tau = \hat{M}(q)(\ddot{q}_d + K_P e + K_D \dot{e}) + \hat{C}(q,\dot{q})\dot{q} + \hat{g}(q) + \hat{B}_v\dot{q}$$

- bad approximate estimates of dynamic coefficients may lead to instability with FBL due to temporary 'non-positive' PD gains (e.g., $\hat{M}(q)K_P < 0$!)
- not easy to turn these laws in adaptive schemes: bounds on PD gains (FFW); inertia inversion/use of acceleration (FBL)

A control law more easily made 'adaptive'



- nonlinear trajectory tracking control (without cancellations) having global asymptotic stabilization properties

(*) $$\tau = \hat{M}(q)\ddot{q}_d + \hat{C}(q,\dot{q})\dot{q}_d + \hat{g}(q) + \hat{B}_v\dot{q}_d + K_P e + K_D \dot{e}$$

- a natural adaptive version would require

$$\dot{\hat{\rho}} =$$ designing a suitable update law (in continuous time)

  - without extra assumptions, it may be possible that joint velocities become eventually "clamped" to those of the desired trajectory (zero velocity error), but a residual position error would be left
  - hard to prove that this may not happen …

- idea: online modification with a reference velocity

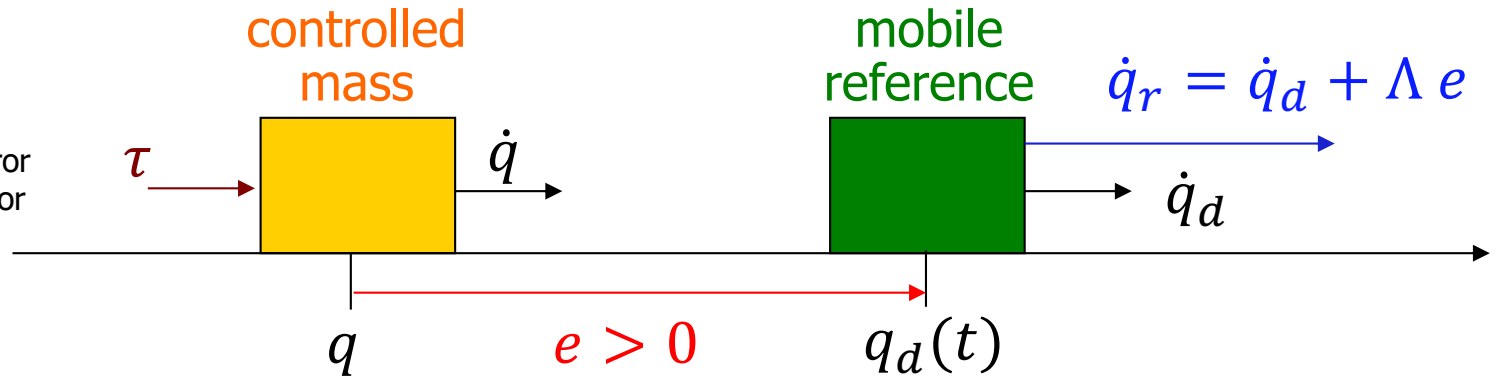
$$\dot{q}_d \quad \longrightarrow \quad \boxed{\dot{q}_r = \dot{q}_d + \Lambda(q_d - q)} \qquad \Lambda > 0$$

typically, $\Lambda = K_D^{-1} K_P$ (all matrices will be chosen diagonal)

Intuitive interpretation of $\dot{q}_r$

- elementary case
  - a mass 'lagging behind' a mobile reference ($e > 0$) at constant speed

$\tau = K_D(\dot{q}_d - \dot{q})$
will zero the velocity error
but not the position error



➡ 'enhanced' velocity error $\sigma = \dot{q}_r - \dot{q} > \dot{q}_d - \dot{q} = \dot{e}$

τ eliminates also
the position error
at steady state

$$\tau = K_D\sigma = K_D(\dot{q}_r - \dot{q}) = K_D(\dot{q}_d + \Lambda\, e - \dot{q}) = K_D\dot{e} + \underbrace{K_D\Lambda}_{K_P}\, e$$

a way to introduce a position feedback in the inner velocity loop
(related to the passivity of the input-output map from τ to $\dot{q}$)

- a mass 'leading in front' of its mobile reference ($e < 0$)

➡ in a symmetric way, a 'reduced' velocity error will appear ($\sigma < \dot{e}$)

Adaptive control law design



- substituting $\dot{q}_r = \dot{q}_d + \Lambda e$, $\ddot{q}_r = \ddot{q}_d + \Lambda \dot{e}$ in the previous nonlinear controller $(*)$ for trajectory tracking

(1)
$$\begin{aligned}\tau &= \hat{M}(q)\ddot{q}_r + \hat{C}(q,\dot{q})\dot{q}_r + \hat{g}(q) + \hat{B}_v\dot{q}_r + K_P e + K_D\dot{e} \\ &= Y(q,\dot{q},\dot{q}_r,\ddot{q}_r)\hat{\rho} \quad + \quad K_P e + K_D\dot{e}\end{aligned}$$

dynamic parametrization of control law using current estimates
(note here the 4 arguments in $Y(\cdot)$!)

PD stabilization
(diagonal matrices, >0)

- update law for the estimates of the dynamic coefficients ($\hat{\rho}$ becomes the r-dimensional state of the dynamic controller)

(2)
$$\dot{\hat{\rho}} = \Gamma Y^T(q,\dot{q},\dot{q}_r,\ddot{q}_r)(\dot{q}_r - \dot{q})$$

σ
modified velocity error

$\Gamma > 0$ (diagonal)

estimation gains
(specifies the variation rate of the estimates)

Asymptotic stability of trajectory error

Theorem

The adaptive controller (1)-(2) makes the tracking error along the desired trajectory globally asymptotically stable

$$e = q_d - q \longrightarrow 0, \; \dot{e} = \dot{q}_d - \dot{q} \longrightarrow 0$$

Proof

- a Lyapunov candidate for the closed-loop system (robot + dynamic controller) is given by

$$V = \frac{1}{2}\sigma^T M(q)\sigma + \frac{1}{2}e^T Q e + \frac{1}{2}\tilde{\rho}^T \Gamma^{-1}\tilde{\rho} \; \geq 0$$

$\sigma = \dot{q}_r - \dot{q} \; (= \dot{e} + \Lambda e)$ modified velocity error

$Q > 0$ constant matrix (to be specified later)

$\tilde{\rho} = \rho - \hat{\rho}$ error in estimation of dynamic coefficients

$$V = 0 \quad \Leftrightarrow \quad \hat{\rho} = \rho, \;\; q = q_d, \;\; \sigma = 0 \;\; (\Rightarrow \; \dot{q} = \dot{q}_d)$$

Proof (cont)



- the time derivative of V is

$$\dot{V} = \frac{1}{2}\sigma^T \dot{M}(q)\sigma + \sigma^T M(q)\dot{\sigma} + e^T Q \dot{e} - \tilde{\rho}^T \Gamma^{-1} \dot{\hat{\rho}}$$

since $\dot{\tilde{\rho}} = -\dot{\hat{\rho}}$ $(\dot{\rho} = 0)$

- the closed-loop dynamics is given by

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) + B_v\dot{q} =$$
$$= \hat{M}(q)\ddot{q}_r + \hat{C}(q,\dot{q})\dot{q}_r + \hat{g}(q) + \hat{B}_v\dot{q}_r + K_P e + K_D \dot{e}$$

subtracting the two sides from $M(q)\ddot{q}_r + C(q,\dot{q})\dot{q}_r + g(q) + B_v\dot{q}_r$ leads to

$$M(q)\dot{\sigma} + (C(q,\dot{q}) + B_v)\sigma =$$
$$= \tilde{M}(q)\ddot{q}_r + \tilde{C}(q,\dot{q})\dot{q}_r + \tilde{g}(q) + \tilde{B}_v\dot{q}_r - K_P e - K_D \dot{e}$$

with $\tilde{M} = M - \hat{M},\ \tilde{C} = C - \hat{C},\ \tilde{g} = g - \hat{g},\ \tilde{B}_v = B_v - \hat{B}_v$

Proof (cont)

- from the property of linearity in the dynamic coefficients, it follows

$$M(q)\dot{\sigma} + (C(q,\dot{q}) + B_v)\sigma = Y(q,\dot{q},\dot{q}_r,\ddot{q}_r)\tilde{\rho} - K_P e - K_D \dot{e}$$

- substituting in $\dot{V}$, together with $\dot{\hat{\rho}} = \Gamma Y^T \sigma$, and using the skew-symmetry of matrix $\dot{M} - 2C$ we obtain

$$\dot{V} = \frac{1}{2}\sigma^T \left(\dot{M}(q) - 2C(q,\dot{q})\right)\sigma - \sigma^T B_v \sigma + \sigma^T Y \tilde{\rho}$$
$$- \sigma^T (K_P e + K_D \dot{e}) + e^T Q \dot{e} - \tilde{\rho}^T Y^T \sigma$$
$$= - \sigma^T B_v \sigma - \sigma^T (K_P e + K_D \dot{e}) + e^T Q \dot{e}$$

- replacing $\sigma = \dot{e} + \Lambda e$ and being $B_v = B_v^T$ (diagonal)

$$\dot{V} = - e^T (\Lambda^T B_v \Lambda + \Lambda^T K_P) e$$
$$- e^T (2\Lambda^T B_v + \Lambda^T K_D + K_P - Q)\dot{e} - \dot{e}^T (B_v + K_D)\dot{e}$$

a complete quadratic form in $e, \dot{e}$!

Proof (end)

- defining now (all matrices are diagonal!)

$$\Lambda = K_D^{-1} K_P > 0 \qquad Q = 2K_P\,(I + K_D^{-1} B_v) > 0$$

cancels the cross-term in $e^T(\dots)\dot{e}$ and leads to

$$\dot{V} = -\,e^T \Lambda^T (B_v + K_D)\Lambda\, e - \dot{e}^T (B_v + K_D)\dot{e}$$

$$= -\,e^T K_P K_D^{-1} (B_v + K_D) K_D^{-1} K_P e - \dot{e}^T (B_v + K_D)\dot{e} \leq 0$$

and thus

$$\dot{V} = 0 \iff e = \dot{e} = 0$$

the thesis follows from Barbalat lemma + LaSalle theorem

⇒ see slide #10 in block 8

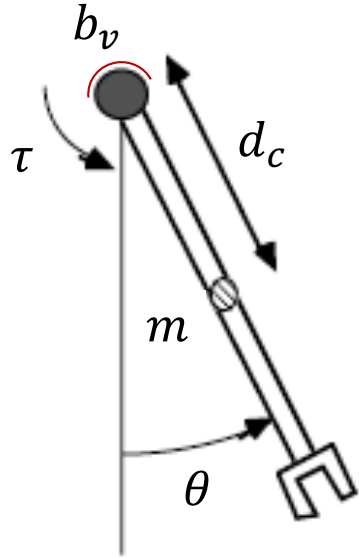
the maximal invariant set $\mathcal{M}$ of states $\subseteq \{\dot{V} = 0\}$ has zero trajectory error ($e = \dot{e} = 0$) and a constant value for $\hat{\rho}$, not necessarily the true one ($\tilde{\rho} \neq 0$)

Remarks



- the adaptive control law (1)-(2) does not require the inverse of the inertia matrix (true or estimated), nor the actual robot acceleration (only the desired acceleration), nor further lower bounds on $K_P > 0, K_D > 0$
- adaptation can also be used for a subset of dynamic coefficients, with the others being known ($Y\rho = Y_{adapt}\,\hat{\rho}_{adapt} + Y_{known}\,\rho_{known}$, adapt only $\hat{\rho}_{adapt}$)
- if the desired trajectory $q_d(t)$ is persistently exciting, then also the estimates of the dynamic coefficients converge to their true values
- condition of persistent excitation
  - for linear systems: # of frequency components in the desired trajectory should be at least twice as large as # of unknown coefficients
  - for nonlinear systems: the condition can be checked only a posteriori
    (a squared integral of motion should always be positively bounded from below)
- a minimal linear parametrization of robot dynamics is not strictly necessary!
  (but choosing dynamic parameters ⇒ coefficients is useful to reduce control complexity)
- the non-adaptive version of (1), with accurate but constant estimates $\hat{\rho}$, is a global tracking controller with static feedback (based on the passivity of robot dynamics)

Case study: Single-link under gravity



model $I\ddot{\theta} + mg_0 d_c \sin\theta + b_v \dot{\theta} = \tau$

linear parametrization

g_0 may also be moved out ...

$$Y(\theta,\dot{\theta},\ddot{\theta})\rho = \begin{pmatrix} \ddot{\theta} & \sin\theta & \dot{\theta} \end{pmatrix} \begin{pmatrix} I \\ mg_0 d_c \\ b_v \end{pmatrix} = \tau$$

adaptive controller

$$e = \theta_d - \theta \qquad \Lambda > 0$$

$$\dot{\theta}_r = \dot{\theta}_d + \frac{k_P}{k_D} e$$

$$\gamma_i > 0, i = 1,2,3$$

$$\tau = \hat{I}\,\ddot{\theta}_r + \widehat{mg_0 d_c} \sin\theta + \hat{b}_v \dot{\theta}_r + k_P e + k_D \dot{e}$$

$$\dot{\hat{\rho}} = \begin{pmatrix} \dot{\hat{I}} \\ \dot{\widehat{mg_0 d_c}} \\ \dot{\hat{b}}_v \end{pmatrix} = \begin{pmatrix} \gamma_1 \ddot{\theta}_r \\ \gamma_2 \sin\theta \\ \gamma_3 \dot{\theta}_r \end{pmatrix} (\dot{\theta}_r - \dot{\theta})$$

Simulation data

- real dynamic coefficients

$$I = 7.5 \qquad mg_0 d_c = 6 \qquad b_v = 1$$

- initial estimates

$$\hat{I} = 5 \qquad \widehat{mg_0 d_c} = 5 \qquad \hat{b}_v = 2$$

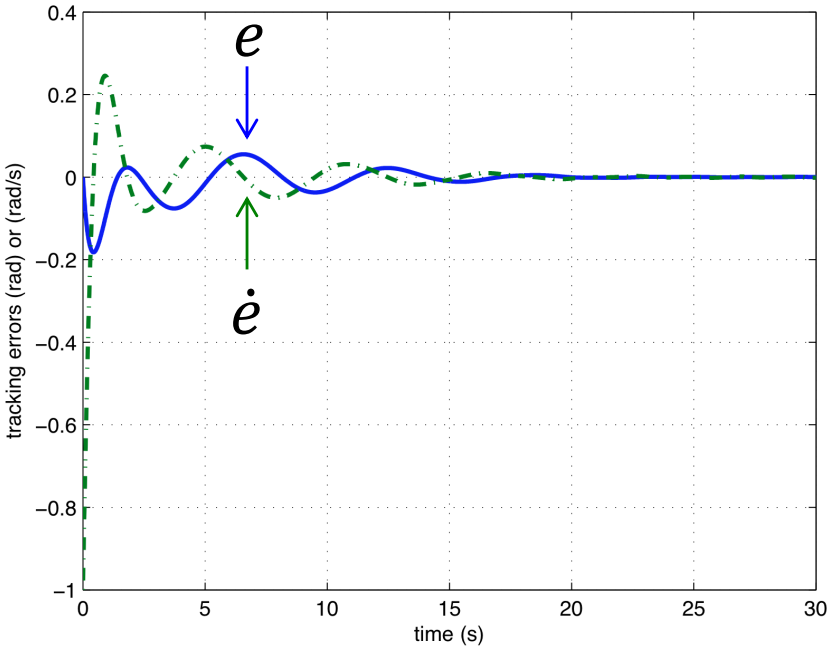
- control parameters

$$k_P = 25 \qquad k_D = 10 \qquad \gamma_i = 5 \quad i = 1,2,3$$

- motion tests with robot starting at $\theta(0) = \dot{\theta}(0) = 0$
  - first trajectory $\theta_d(t) = -\sin t$
  - second trajectory

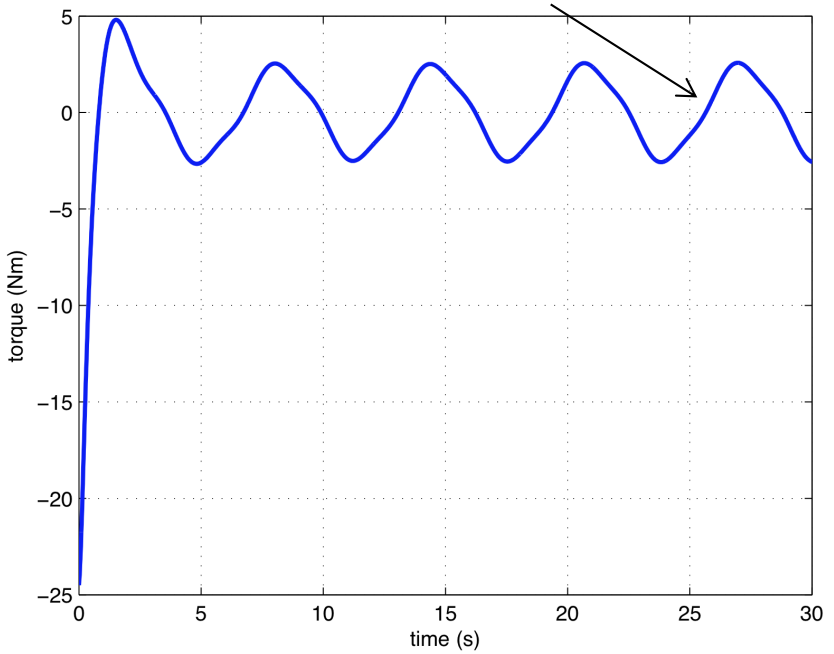
$\ddot{\theta}_d(t) =$ (periodic) bang-bang acceleration profile with $A = 1$ rad/s^2, $\omega = 1$ rad/s

Results
first trajectory



position and velocity errors

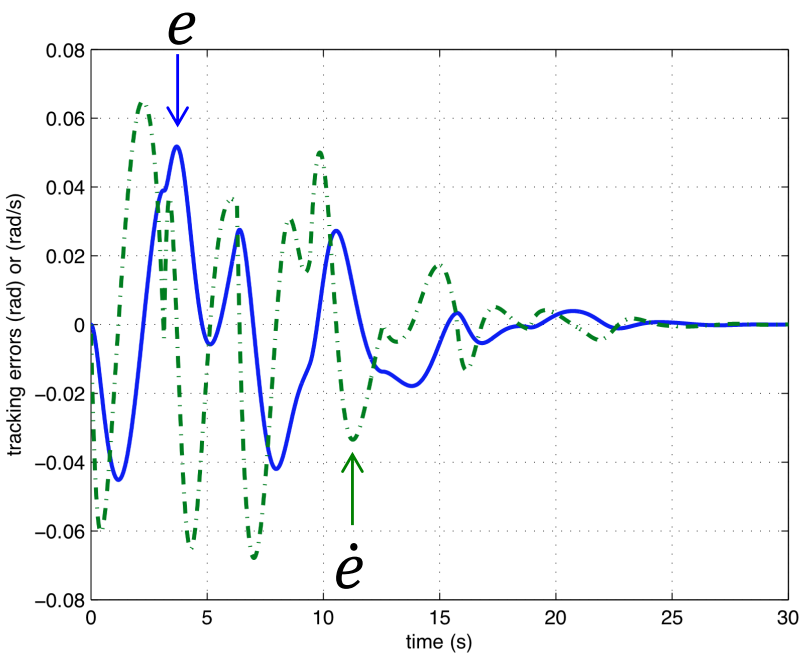
no sinusoidal regime at steady state
(due to nonlinear system dynamics!)



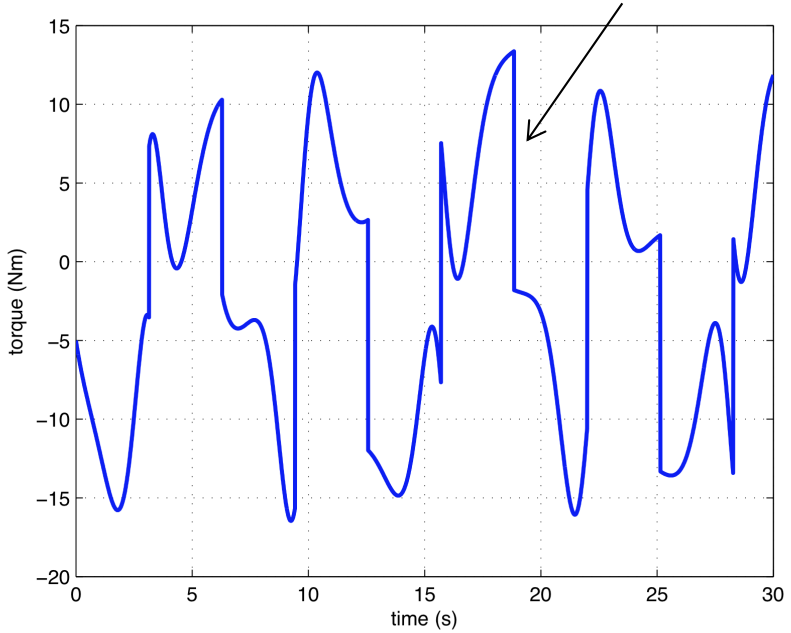
control torque

$$\theta_d(t) = -\sin t$$

Results
second trajectory



torque discontinuities are due to those of the desired acceleration

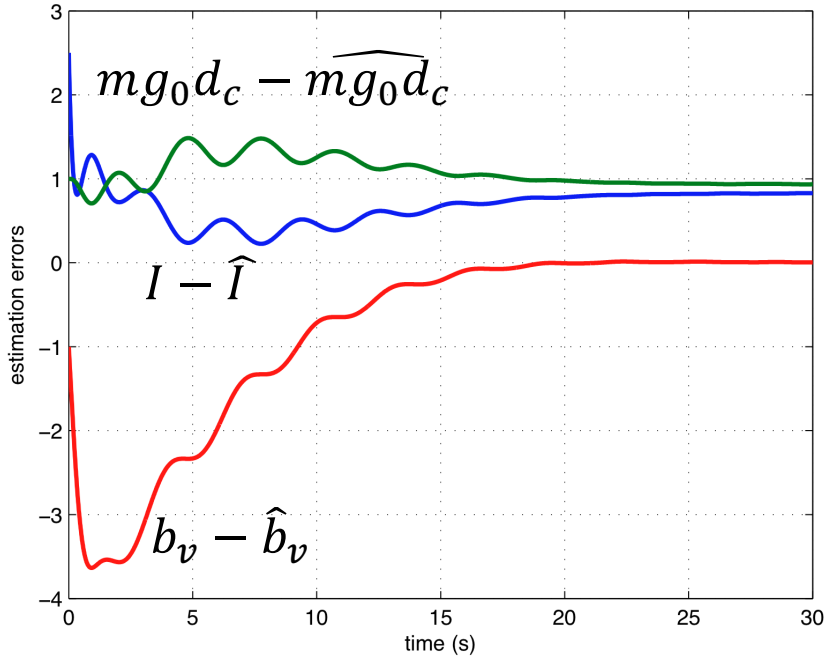


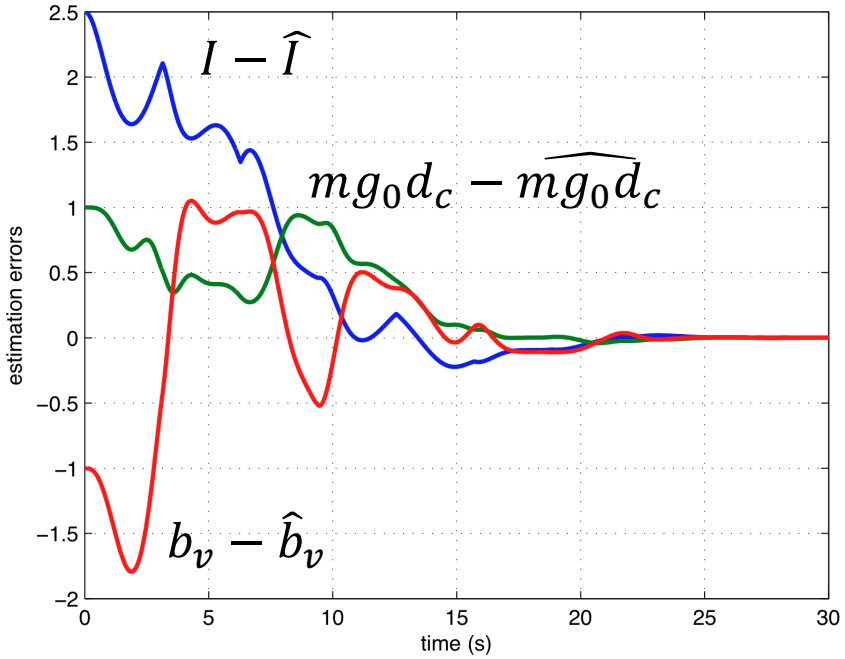
position and velocity errors

control torque

$\ddot{\theta}_d(t)$ = (periodic) bang-bang acceleration profile

Estimates of dynamic coefficients





errors $\tilde{\rho} = \rho - \hat{\rho}$

first trajectory

only the estimate of the viscous friction coefficient converges to the true value

second trajectory

all three estimates of the dynamic coefficients converge to their true values

Adaptive regulation
a special case



- adaptation in case q_d is constant
- no simplifications of the adaptive control law (1)-(2), which is designed in fact for the general tracking case …

$$\tau = \hat{M}(q)\ddot{q}_r + \hat{C}(q,\dot{q})\dot{q}_r + \hat{g}(q) + \hat{B}_v\,\dot{q}_r + K_P e + K_D \dot{e}$$

$$\dot{\hat{\rho}} = \Gamma\, Y^T(q,\dot{q},\dot{q}_r,\ddot{q}_r)(\dot{q}_r - \dot{q})$$

since $\dot{q}_r = \Lambda(q_d - q)$ and $\ddot{q}_r = -\Lambda\dot{q}$ do not vanish!

- a different case would be the availability of an adaptive version of the trajectory tracking controller

$$\tau = \hat{M}(q)\ddot{q}_d + \hat{C}(q,\dot{q})\dot{q}_d + \hat{g}(q) + \hat{B}_v\,\dot{q}_d + K_P e + K_D \dot{e}$$

since, when q_d collapses to a constant, only adaptation of the gravity term is left over (as we would naturally expect…)

An efficient adaptive regulator



- use a linear parametrization of the gravity term only

$$g(q) = G(q)\rho_g$$

with a r_g-dimensional vector ρ_g

- an adaptive regulator yielding global asymptotic stability of the equilibrium state $(q_d, 0)$ is provided by

$$\tau = G(q)\hat{\rho}_g + K_P(q_d - q) - K_D\dot{q}$$

$$\dot{\hat{\rho}}_g = \gamma G^T(q)\left(\frac{2e}{1 + 2\|e\|^2} - \beta\dot{q}\right), \qquad \gamma > 0$$

where $e = q_d - q$, $K_P > 0$, $K_D > 0$ (symmetric), and $\beta > 0$ is chosen sufficiently large

for the (complex) analysis, see P. Tomei, IEEE TRA, 1991, available as extra material on the course website

An adaptive regulator

Sketch of asymptotic stability analysis



- use the function

$$V = \frac{\beta}{2}(\dot{q}^T M(q)\dot{q} + e^T K_P e) - \frac{2\dot{q}^T M(q) e}{1 + 2\|e\|^2} + \frac{1}{2}\left(\hat{\rho}_g - \rho_g\right)^T \left(\hat{\rho}_g - \rho_g\right)$$

- a sufficient condition for V to be a Lyapunov candidate is that

$$\beta > \frac{2M_M}{\sqrt{M_m K_{P,m}}}$$

- a sufficient condition which guarantees also that

$$\dot{V} = \cdots \leq -a\|e\|^2 - b\|\dot{q}\|^2 \leq 0, \qquad a > 0, b > 0$$

is

$$\beta > \max\left\{\frac{2M_M}{\sqrt{M_m K_{P,m}}}, \frac{1}{K_{D,m}}\left(\frac{K_{D,m}^2}{2K_{P,m}} + 4M_M + \frac{\alpha_C}{\sqrt{2}}\right)\right\}$$

$\|C(q,\dot{q})\| \leq \alpha_C \dot{q}$

memo: for any symmetric, positive definite matrix A

$A_M = \lambda_{\max}(A) = \sqrt{\lambda_{\max}(A^T A)} = \|A\|$

$A_m = \lambda_{\min}(A)$

and thus, e.g., $\frac{1}{2}\dot{q}^T M(q)\dot{q} \geq \frac{1}{2} M_m \|\dot{q}\|^2$